

1. Largest valid value of r :

Example of win records:

Justification:

Larger values of r wouldn't work because ...

2. Simplify the following expression, making one modification per line.

On each line you should also briefly explain what rule was used to get that line from the one above.

$$\neg(\neg(p \wedge q) \vee \neg(\neg q \wedge r))$$

1. $\neg((\neg p \vee \neg q) \vee \neg(\neg q \wedge r)) \rightarrow$ Demorgan's on $\neg(p \wedge q)$

2. $\neg((\neg p \vee \neg q) \vee (q \vee \neg r)) \rightarrow$ Demorgan's on $\neg(\neg q \wedge r)$

3. $\neg(\neg p \vee \neg q \vee q \vee \neg r) \rightarrow$ Associative Law

4. $p \wedge q \wedge \neg q \wedge r \rightarrow$ Demorgan's on outer parenthesis

5. $p \wedge F \wedge r \rightarrow q \wedge \neg q$ annihilates to F

6. $F \rightarrow p \wedge F \wedge r$ is always F

3. Definition: $aXb = \neg(a \vee b)$

a) $\neg p$

$\leftrightarrow$

$\leftrightarrow$

$\leftrightarrow$

b) $p \vee q$

...

c) $p \wedge q$

...